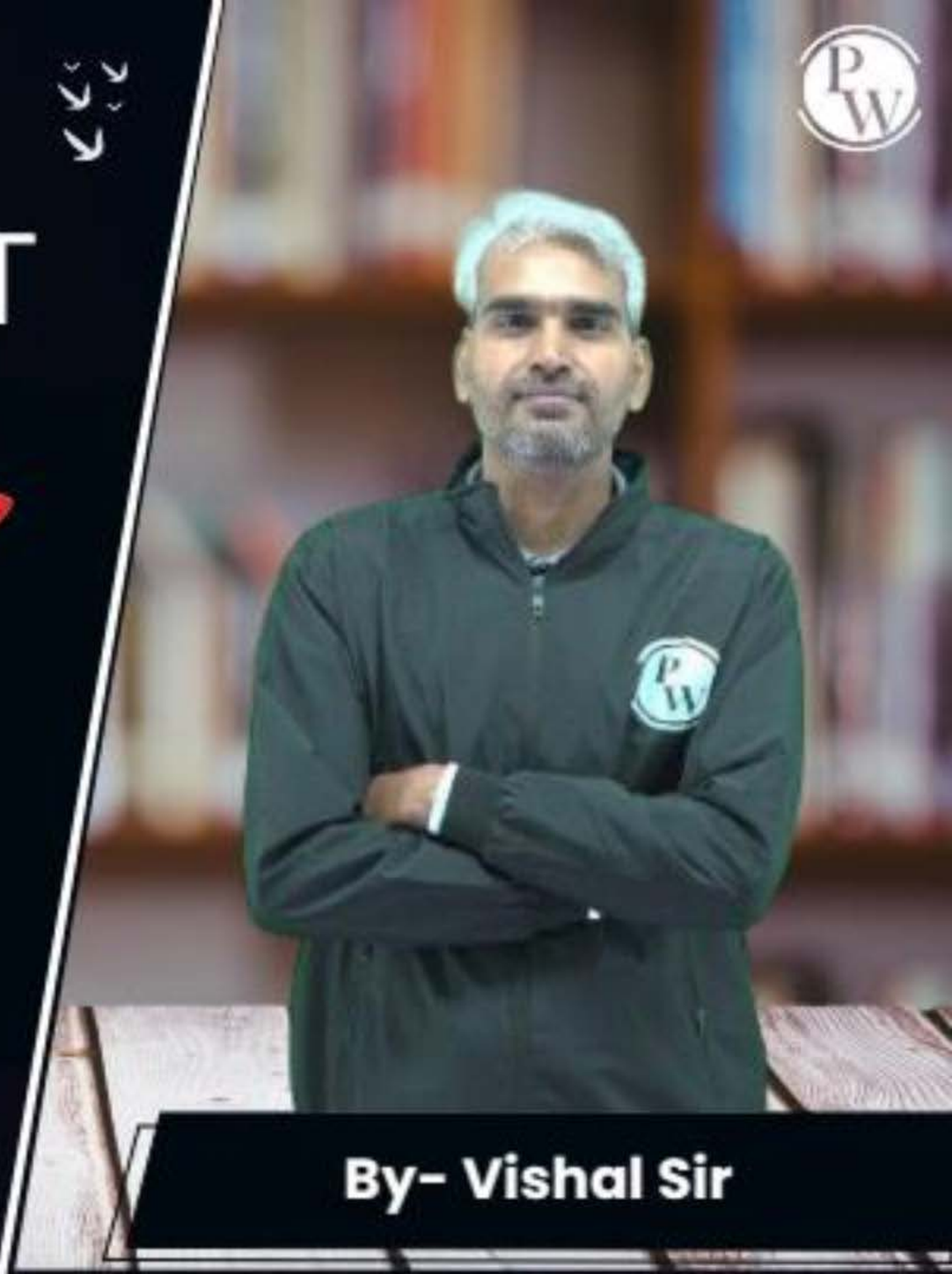
Computer Science & IT



Discrete Mathematics

Set Theory & Algebra

Lecture No. 16



By- Vishal Sir

Recap of Previous Lecture

Topic

Practice questions



Slide

Topics to be Covered



- Topic: Bounded lattice
- Topic: Complement of an element in a lattice
- Topic: Complemented lattice
- Topic: Sub lattice
- Topic: Distributive lattice
- Topic: Boolean algebra

Slide

Bounded Lattice

→ A lattice in which both minimum as well as maximum element exist is called a bounded lattice



Topic : Bounded Lattice

✓ Let $[L, \vee, \wedge]$ be a lattice

① If there exists an element $I \in L$ s.t. $a \vee I = I \quad \forall a \in L$.
then 'I' is called Universal upper bound
(or) Upper bound of lattice L

* "I" is actually the maximum element of lattice.

② If there exist an element $O \in L$.
such that, $a \wedge O = O \quad \forall a \in L$
then, 'O' is called Universal lower bound
(or) lower bound of lattice L

* 'O' is actually the minimum element

* A lattice $[L, \vee, \wedge]$ is called a bounded lattice if and only if both 'I' and 'O' exist in the lattice
(or)
* A lattice is called a bounded lattice if and only if both maximum & minimum element exist.

Slide

Note :- ① Every finite lattice is a bounded lattice

② If lattice is an infinite lattice then it may or may not be a bounded lattice

eg: $(Z, \leq)$ and $(N, \leq)$ are example of unbounded lattice where both Z & N are infinite sets

Bounded lattice w.r.t. infinite set

eg: let $A = \{x \mid x \in R \text{ and } 0 \leq x \leq 1\}$

↳ A is an infinite set.

We know $(A, \leq)$ is a lattice where 'A' is an infinite set

but 0 = Minimum element

& 1 = Maximum element

→ Both minimum & maximum element exist so it is a bounded lattice.



Topic : Complement of an element in a lattice

Complement of an element of lattice $[L, \vee, \wedge]$ exists only if lattice is a bounded lattice

P only if Q
|||
if P then Q
|||
$P \longrightarrow Q$

i.e., ① If Complement exists for any element of lattice $[L, \vee, \wedge]$, then it is a bounded lattice

& ② If lattice is not a bounded lattice then Complement does not exist for any element of that lattice

Slide



Topic : Complement of an element in a lattice

Let $[L, \vee, \wedge]$ be a bounded lattice,

Where I = Universal upper bound (i.e. Maximum element of lattice L)

& O = Universal lower bound (i.e., Minimum element of lattice L)

For an element $a \in L$

if there exist some elements $b \in L$

such that (i) $a \vee b = I$ (maximum element)

$lub(a, b) = I$

and (ii) $a \wedge b = O$ (minimum element)

$glb(a, b) = O$

then element a & b are called Complement of each other w.r.t lattice $[L, \vee, \wedge]$

Slide

Note:

① In a bounded lattice,
'I' and '0' are always Complement of each other
and they do not have any other Complement.

i.e.: The only Complement of Maximum element(I) is minimum element(0)
and The only Complement of minimum element(0) is maximum element(I)

② In a bounded lattice,
Complement need not exist for every element
of the lattice, and if exists it need not be unique
{i.e. some elements may have more
than one Complement

↳ In a bounded lattice number of Complements of an element
of the lattice may be '0' or '1' or more



Topic : Complemented lattice

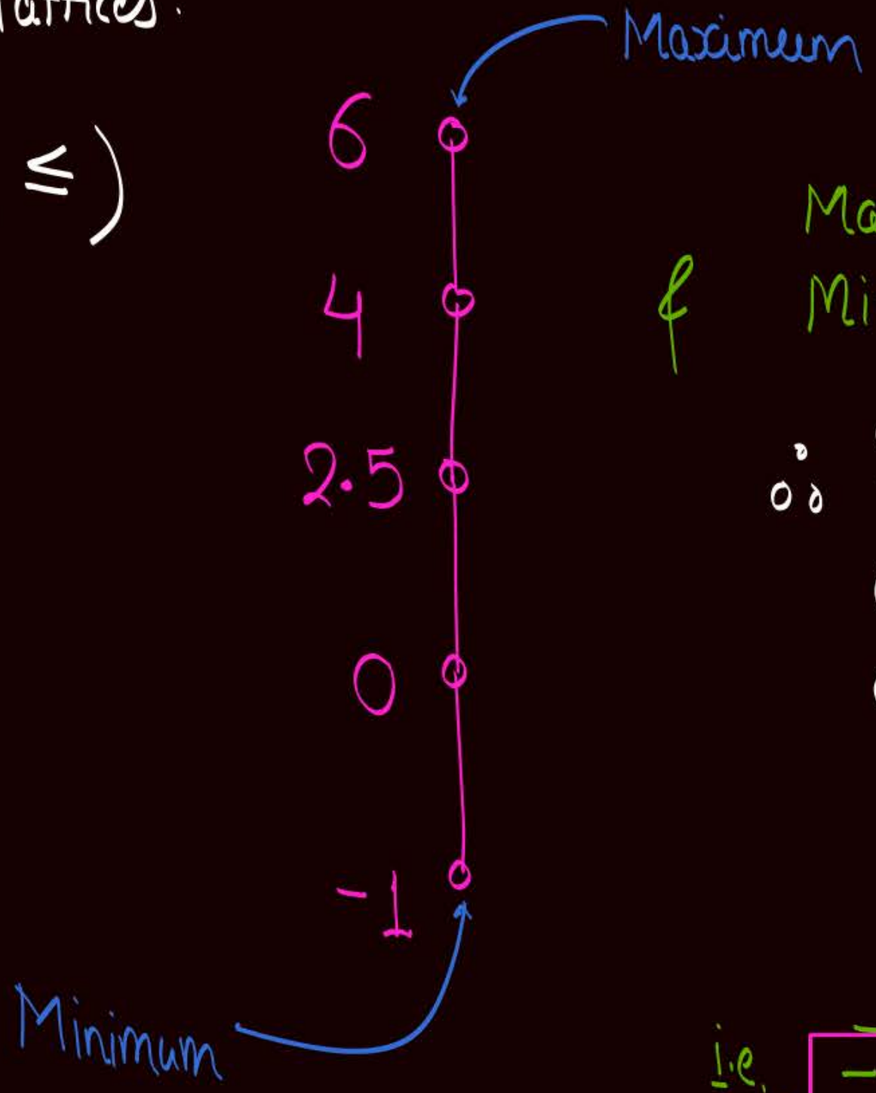
A lattice is called a Complemented lattice if and only if at least one Complement exist for every element of the lattice.

- i.e. in a Complemented lattice every element will have '1' or "more" Complements.

Slide

Find the complement/s of Every element (if Exist) of the following lattices.

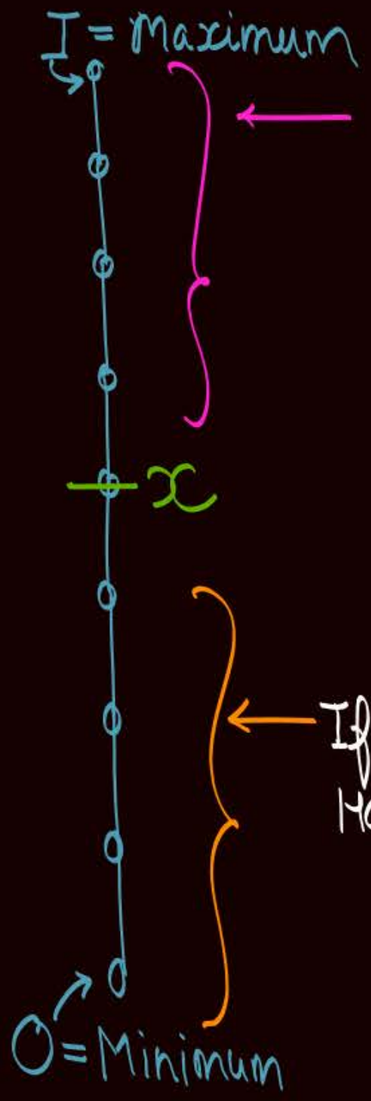
(i) $(\{-1, 0, 2.5, 4, 6\}, \leq)$



Maximum element = 6
&
Minimum element = -1

∴ '-1' & '6' are definitely Complement of each other
&
No other Complement will exist for them

i.e., $\overline{-1} = 6$ & $\overline{6} = -1$



If we choose any element 'a' above 'x' in the Hasse diagram, then $glb(x, a) = x \neq$ Minimum

∴ No such element 'a' can be complement of 'x'

$x \vee x = x \neq$ Maximum

$x \wedge x = x \neq$ Minimum

} ∴ x can not be the complement of 'x' itself.

∴ $\bar{x}$ can not exist

If we choose any element 'b' below 'x' in the Hasse diagram, then $lub(x, b) = x \neq$ Maximum

∴ No such element 'b' can be complement of 'x'

Note:- ① A TOSET is always a lattice, but Every lattice need not be a TOSET.

② If TOSET is a bounded lattice, then

Complement of minimum element = maximum element

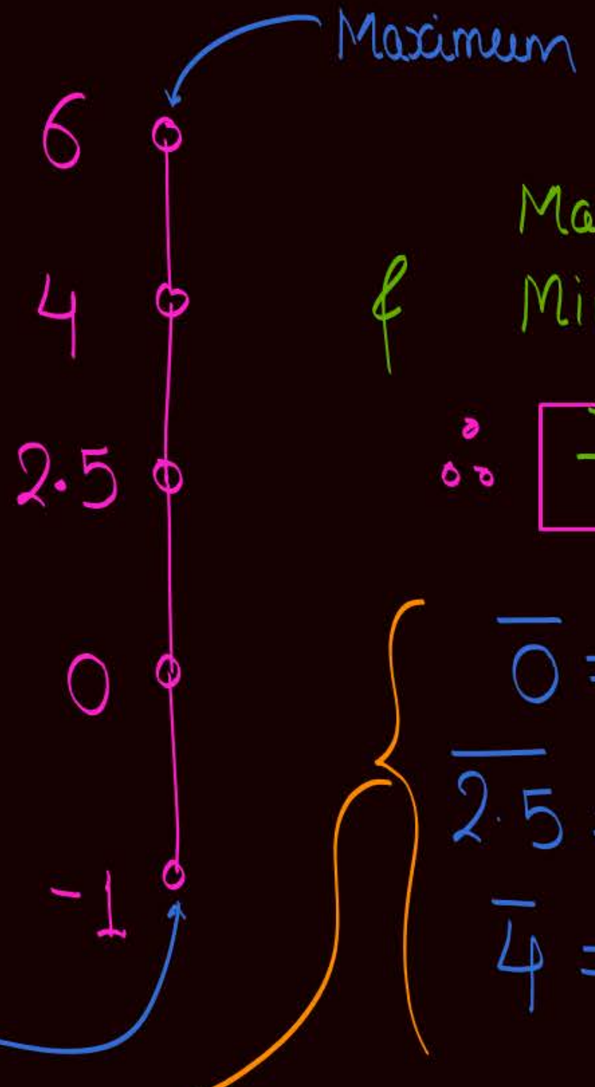
and Complement of maximum element = minimum element

but Complement does not exist for any other element in a lattice w.r.t. TOSET

③ If two elements lie on the same line in a Hasse diagram, then they can not be Complement of each other. Except for minimum & maximum element

Find the complements of every element (if exist) of the following lattices.

(i) $(\{-1, 0, 2.5, 4, 6\}, \leq)$



Maximum element = 6
&
Minimum element = -1

$\therefore \overline{-1} = 6$ & $\overline{6} = -1$

$\overline{0}$ = DNE

$\overline{2.5}$ = DNE

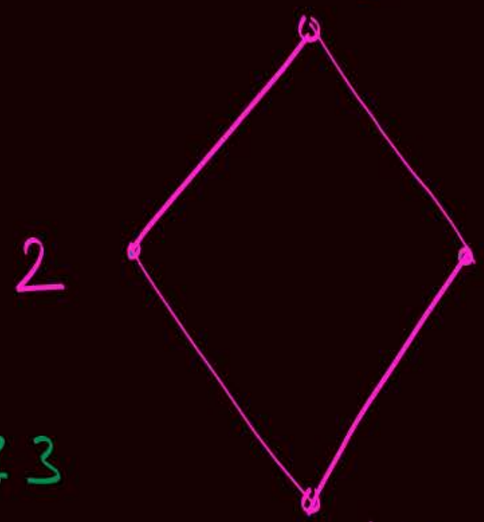
$\overline{4}$ = DNE

∴ Given lattice is not a Complemented lattice.

Find the complement of Every element (if Exist) of the following lattices.

(ii) $(D_6, \div)$

$\{1, 2, 3, 6\}$



$lub(2,3) = 6$ = Maximum

&

$glb(2,3) = 1$ = Minimum

∴ 2 & 3 are Complement of each other

$\bar{1} = 6$

$\bar{6} = 1$

$\bar{2} = 3$

$\bar{3} = 2$

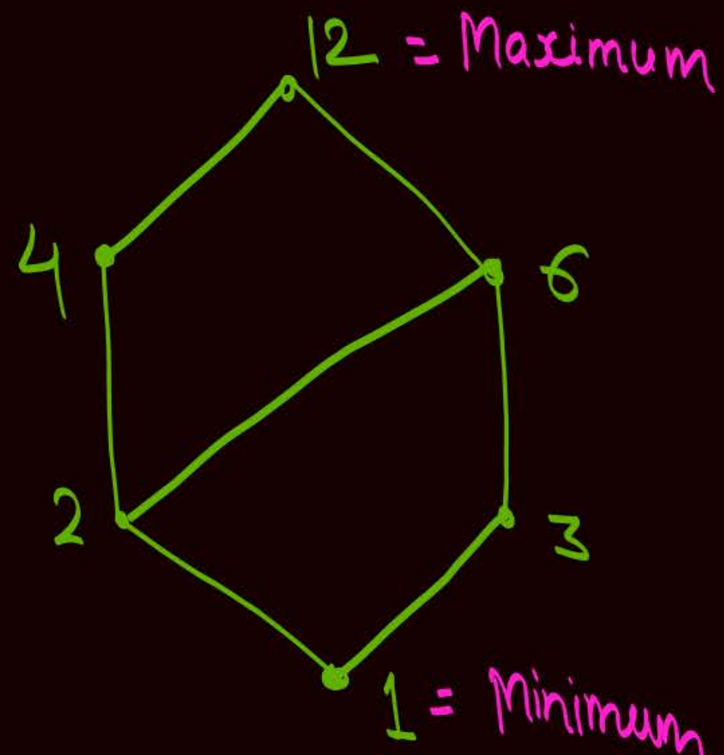
Every element has Exactly '1' complement

⇓

∴ Complemented lattice.

Find the complement of Every element (if Exist) of the following lattices.

(iii) $(D_{12}, \div)$



$\bar{1} = 12$

$\bar{12} = 1$

$\bar{2}$ = Does not exist

$\bar{3} = 4$

$\bar{4} = 3$

$\bar{6}$ = Does not exist

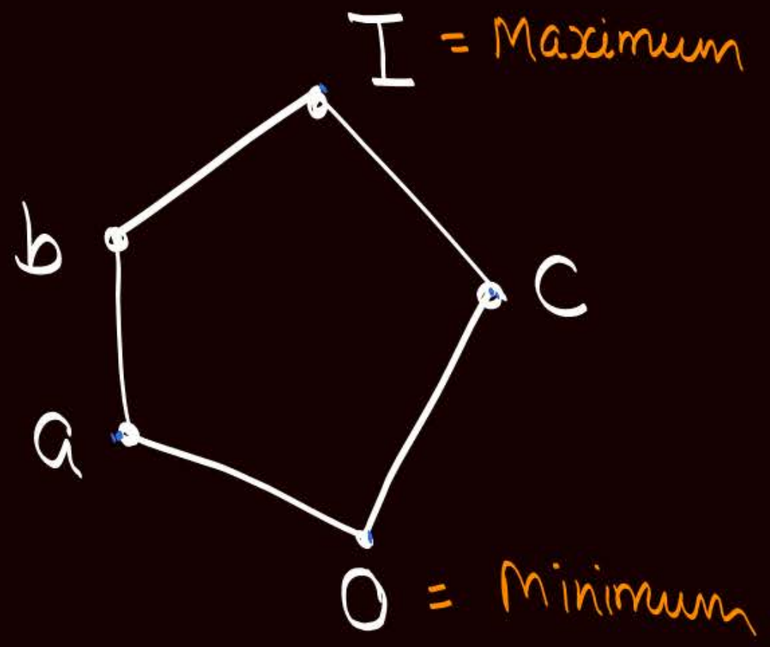
∴ Not a Complemented lattice.

lub(2,3) = 6 ≠ Maximum
∴ 2 & 3 are not Complement

lub(3,4) = 12 = Max
glb(3,4) = 1 = Min
∴ 3 & 4 are Complement of each other

glb(4,6) = 2 ≠ Minimum
∴ 4 & 6 are not Complement

Eg:- Consider the following POSET diagram for a lattice, find the complements for every element.



$\bar{O} = I$

$\bar{I} = O$

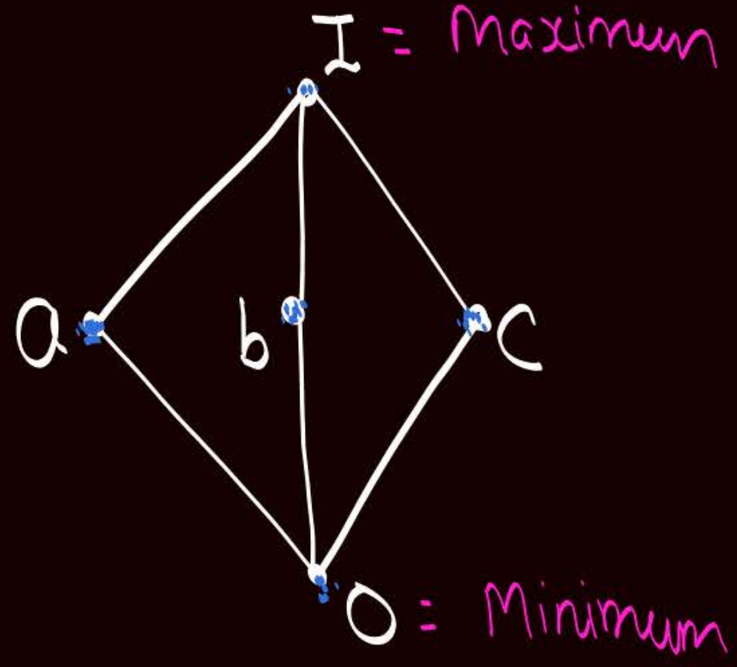
$\bar{a} = c$

$\bar{b} = c$

$\bar{c} = a, b$

∴ it is a Complemented lattice

Eg:- Consider the following POSET diagram for a lattice, find the complements for every element.



$\bar{O} = I$

$\bar{I} = O$

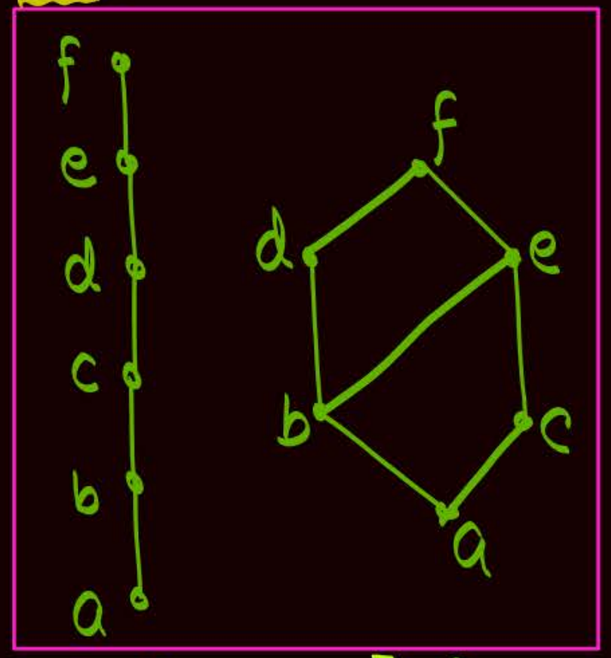
$\bar{a} = b, c$

$\bar{b} = a, c$

$\bar{c} = a, b$

∴ it is a Complemented lattice

Note:-

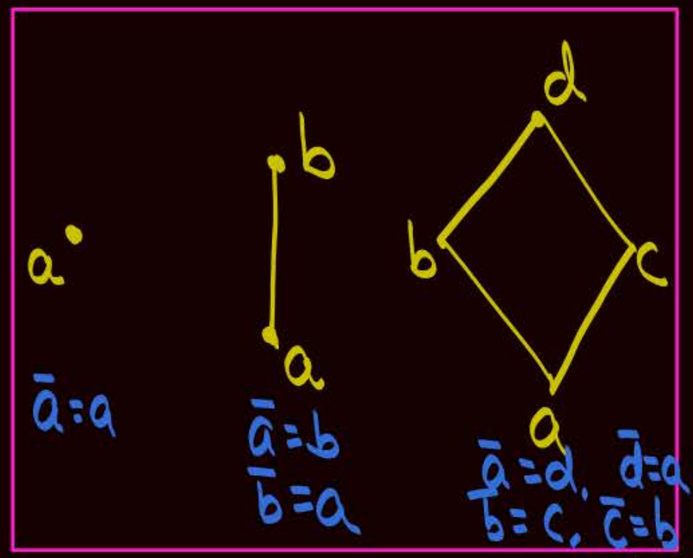


$\bar{a} = f$
$\bar{f} = a$

Complement does not exist for any other element

$\bar{a} = f$
$\bar{f} = a$
$\bar{d} = c$
$\bar{c} = d$
$\bar{b} = DNE$
$\bar{e} = DNE$

These are not Complemented lattice

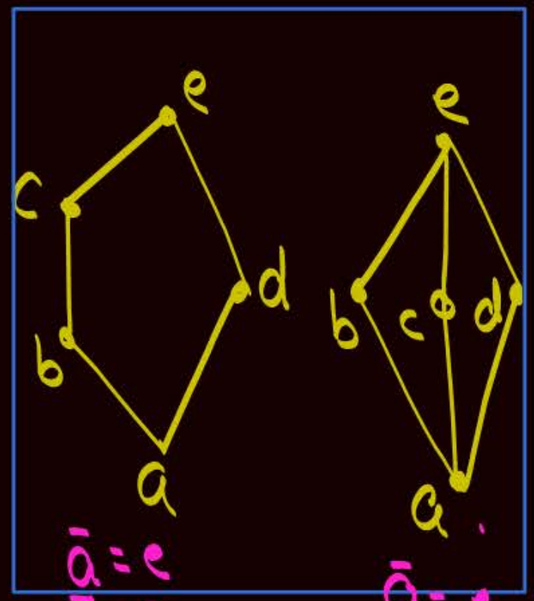


$\bar{a} = a$

$\bar{a} = b$
$\bar{b} = a$

$\bar{a} = d,\ \bar{d} = a$
$\bar{b} = c,\ \bar{c} = b$

Every element has Exactly '1' Complement



$\bar{a} = e$
$\bar{b} = a, d$
$\bar{c} = d$
$\bar{d} = b, c$

$\bar{a} = e$
$\bar{b} = a$
$\bar{c} = c, d$
$\bar{d} = b, d$
$\bar{e} = b, c$

All are Complemented lattice



Topic : Sublattice

Let $[L, \vee, \wedge]$ be a lattice, and M be any subset of set L

'M' is a sub-lattice of lattice 'L' if and only if,

(i) $[M, \vee, \wedge]$ is a lattice

{ i.e. lub as well as glb must exist for Every pair of elements of set M in set M itself }

and (ii) lub and glb for Every pair of elements in lattice $[M, \vee, \wedge]$ must be Exactly same as lub and glb for that pair of elements in lattice $[L, \vee, \wedge]$

i.e. $\forall a, b \in M$

$\text{lub}(a,b)$ in M = $\text{lub}(a,b)$ in L

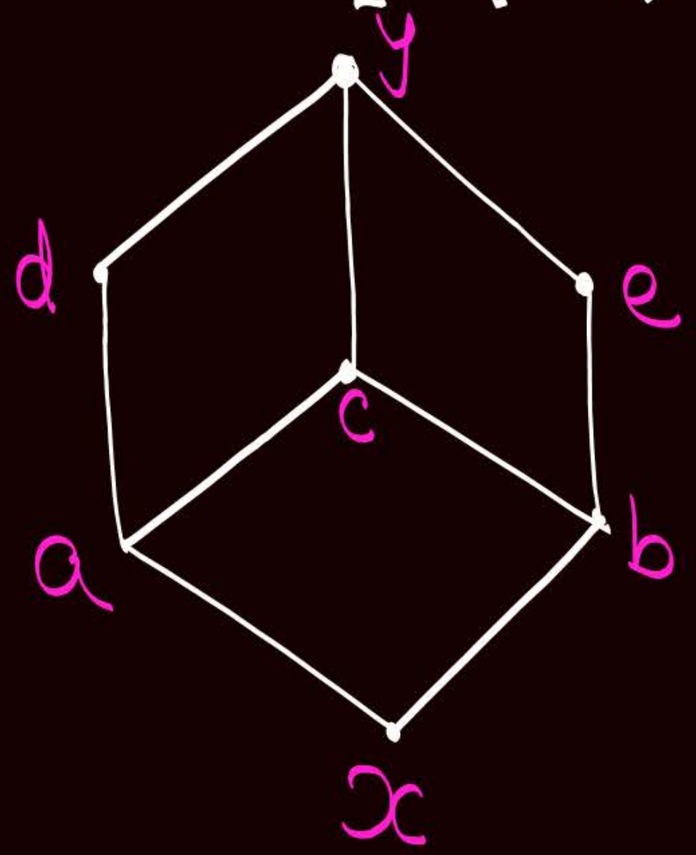
&

$\text{glb}(a,b)$ in M = $\text{glb}(a,b)$ in L

Slide

eg Consider the following Lattice 'L'

"L" = {x, y, a, b, c, d, e}



Which of the following is/are Sublattice of lattice L

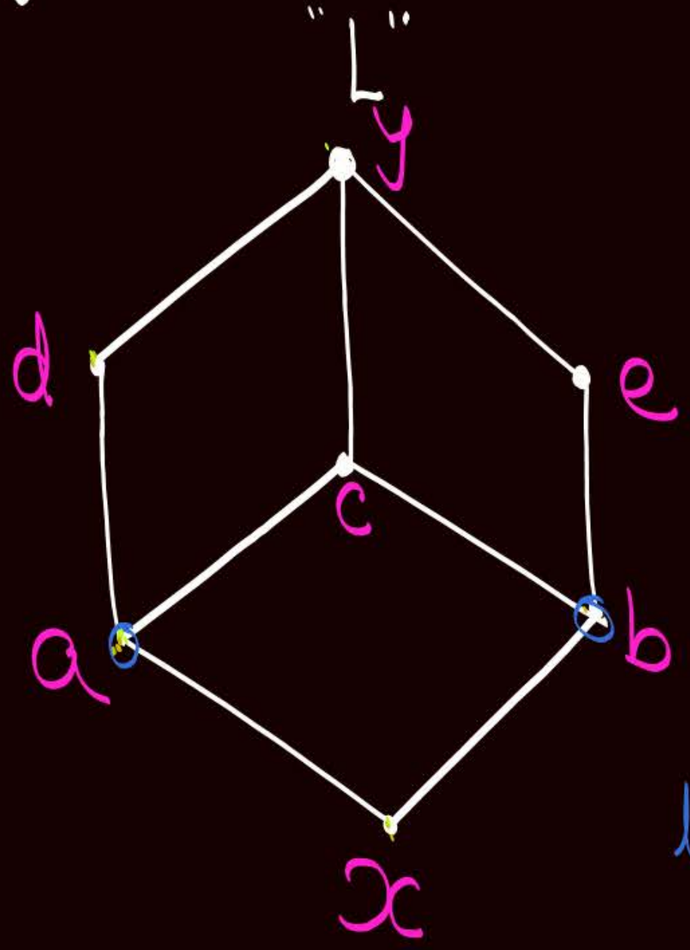
(i) $L_1 = \{x, a, b, y\}$

(ii) $L_2 = \{x, c, d, y\}$

(iii) $L_3 = \{x, a, c, y\}$

(iv) $L_4 = \{x, a, e, y\}$

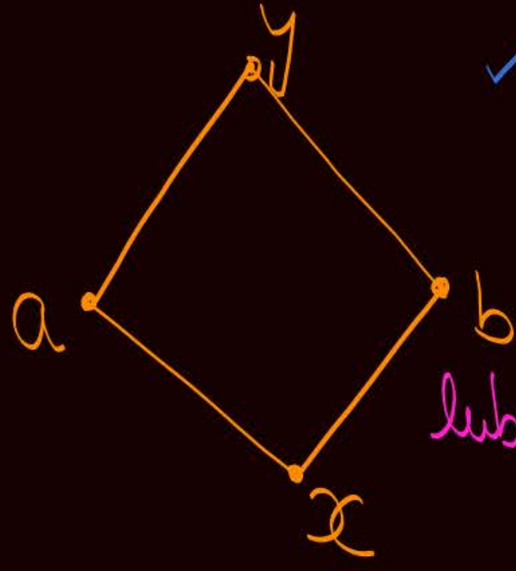
eg Consider the following lattice 'L'



lub(a,b) = c

Which of the following is/are sublattices of lattice L

(i) $L_1 = \{x, a, b, y\}$



L_1

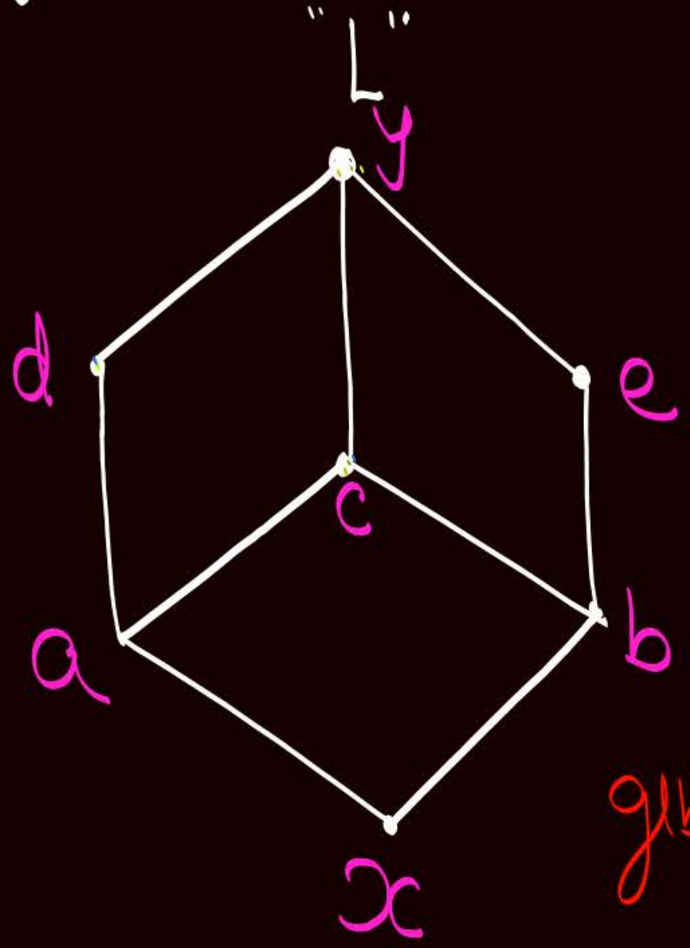
✓ ① It is a lattice

② But

lub(a,b) in L_1 = y ≠ lub(a,b) in L

∴ L_1 is not a sub-lattice of L

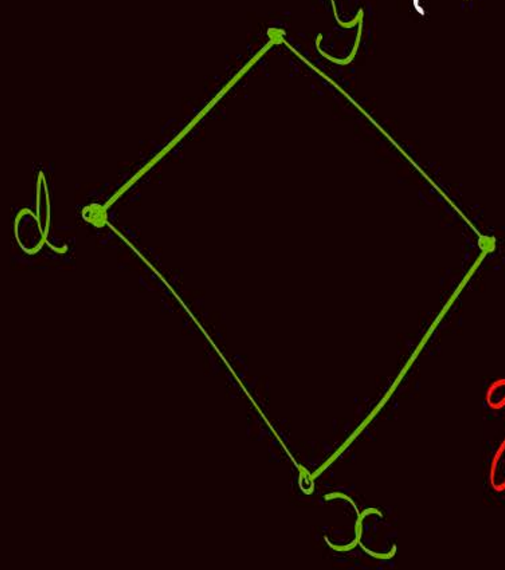
eg Consider the following Lattice 'L'



glb(c,d) = a

Which of the following is/are sublattices of lattice L

(ii) $L_2 = \{x, c, d, y\}$



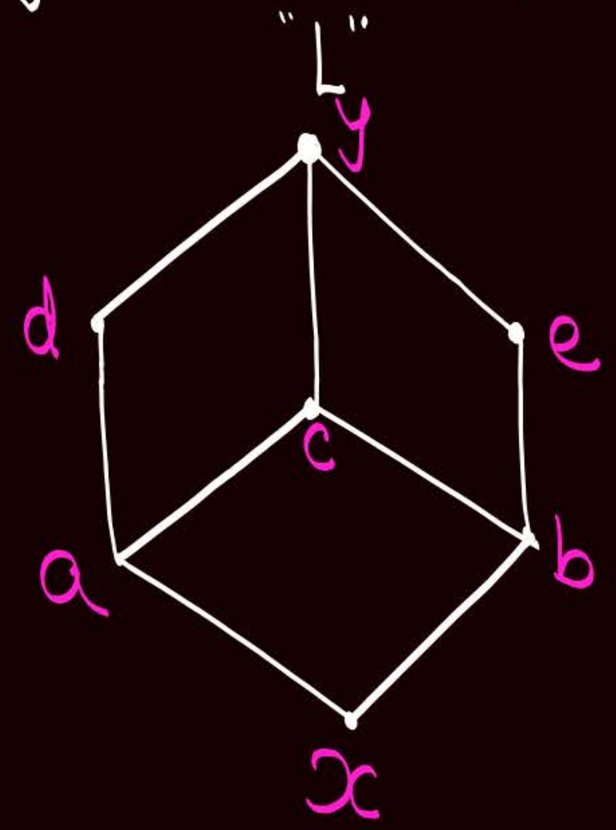
① It is a lattice

② But

glb(c,d) in L_2 = x ≠ glb(c,d) in L

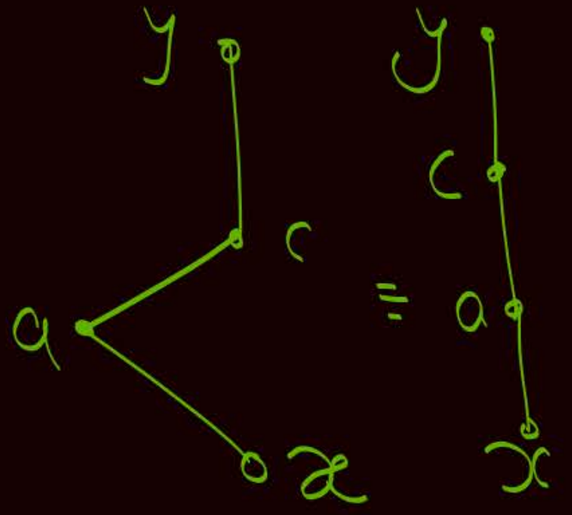
∴ Not a Sub-lattice of L

eg Consider the following lattice 'L'



Which of the following is/are sublattices of lattice L

(iii) $L_3 = \{x, a, c, y\}$

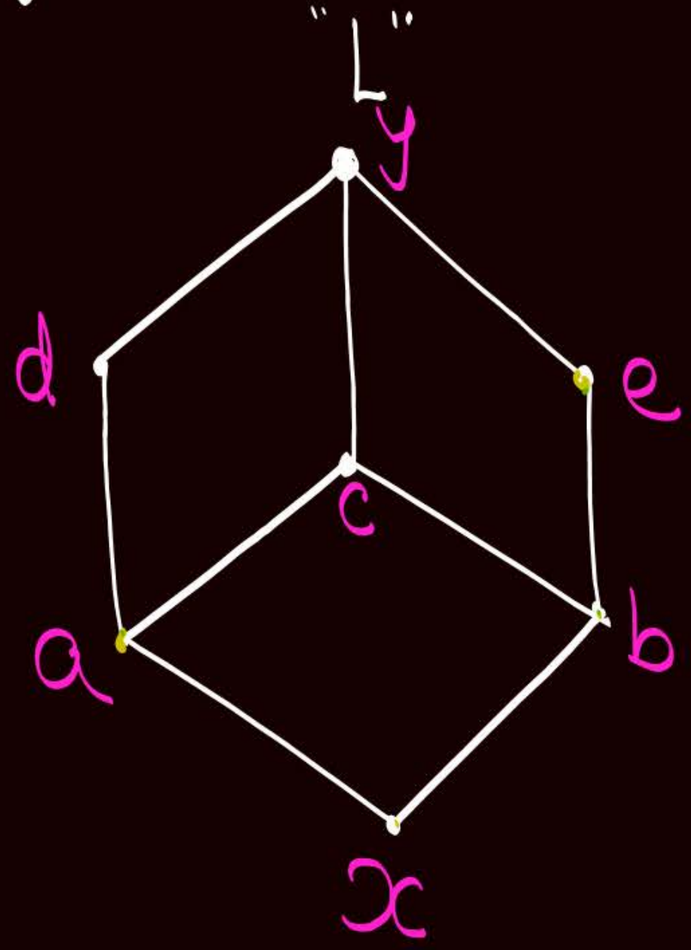


① It is a lattice

② lub & glb for every pair of elements in L_3 will be same as lub & glb for that pair of elements in L

∴ L_3 is a sub-lattice of L

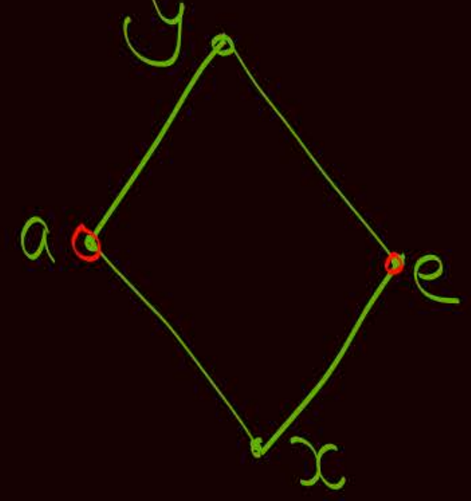
eg Consider the following Lattice 'L'



lub(a,e) = y
glb(a,e) = x

Which of the following is/are sublattices of lattice L

(iv) $L_4 = \{x, a, e, y\}$



① It is a lattice

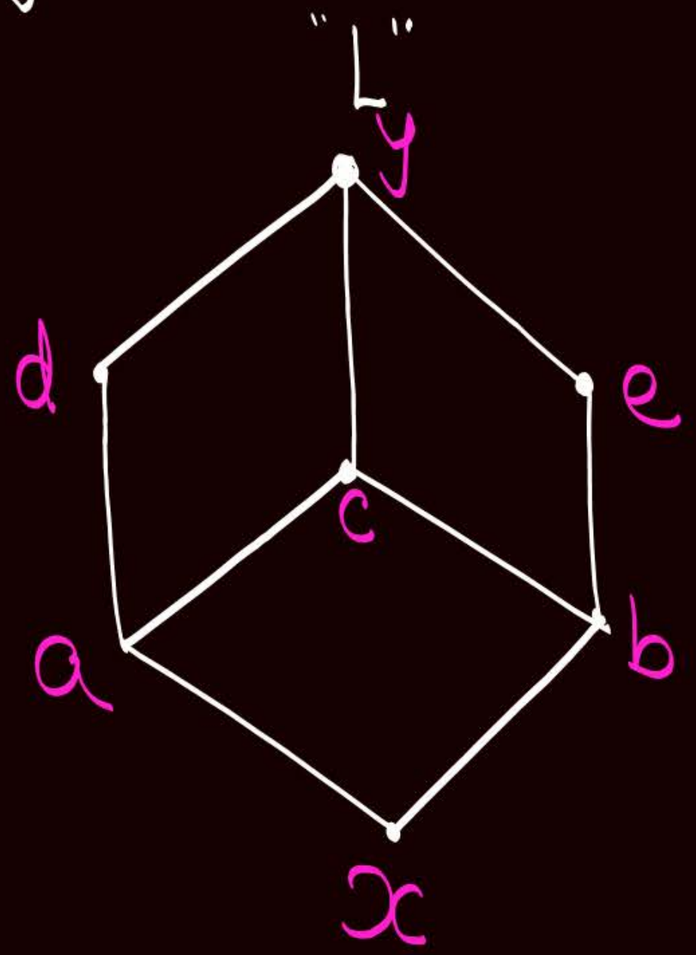
② $\underset{\text{in } L_4}{lub(a,e)} = y = lub(a,b) \text{ in } L$

&

$\underset{\text{in } L_4}{glb(a,e)} = x = glb(a,e) \text{ in } L$

$\therefore L_4$ is sub-lattice of L

eg Consider the following lattice 'L'



Which of the following is/are sublattices of lattice L

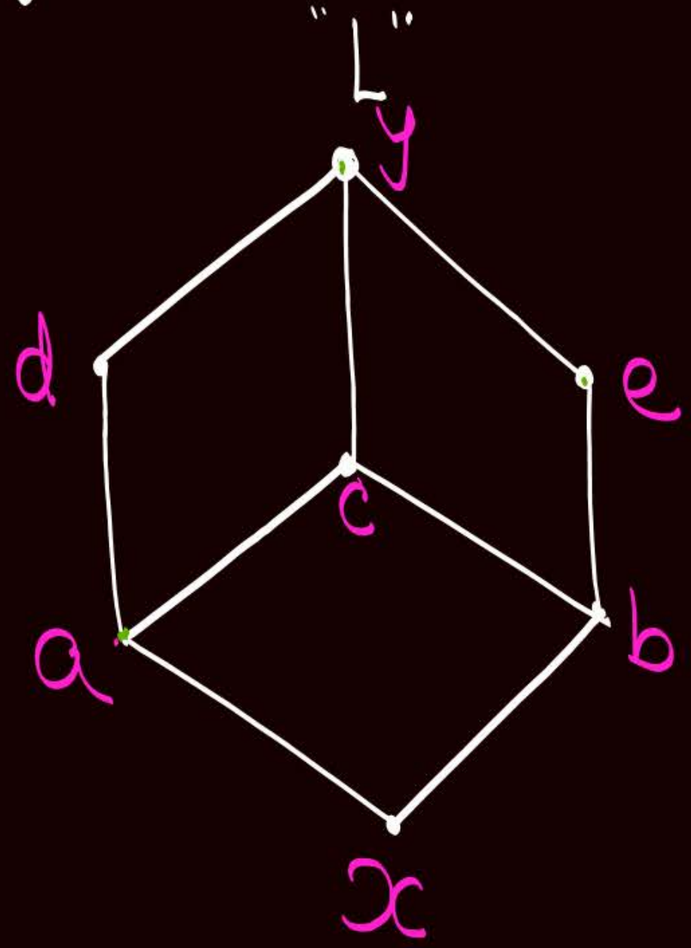
(i) $L_1 = \{x, a, b, y\}$

(ii) $L_2 = \{x, c, d, y\}$

(iii) $L_3 = \{x, a, c, y\}$

(iv) $L_4 = \{x, a, e, y\}$

eg Consider the following lattice 'L'



Which of the following is/are sub-lattices of lattice L

(a) $L_1 = \{a, e, y\}$ =



← it is not a lattice

(b) $L_2 = \{a, b, c\}$ =



← it is not a lattice

(c) $L_3 = \{x, a, d\}$



← it is a lattice ✓
← on the same line so lub & glb also same

(d) $L_4 = \{a, c, y\}$



← it is a lattice ✓
← on the same line so lub & glb also same



2 mins Summary

- Topic: Bounded lattice
- Topic: Complement of an element in a lattice
- Topic: Complemented lattice
- Topic: Sub lattice
- Topic: Distributive lattice
- Topic: Boolean algebra

THANK - YOU